



Interpretability from the Ground Up: Stakeholder-Centric Design of Automated Scoring in Educational Assessments

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Background

- Motivation:** Automated scoring (AS) is rapid and scalable, but stakeholder trust is crucial for adoption in high-stakes educational assessments. Interpretable AS helps foster that trust.
- Research-to-Practice Gap:** Despite demand, interpretable AS remains limited in practice.
- Illustrative Example:** 2023 NAEP Math Automated Scoring Challenge winners achieved human-like scoring accuracy, but *none of the* submissions met the interpretability criteria.
- Underlying Issue:** We attribute this gap to the lack of a stakeholder need-finding process.

Contributions

Problem Formulation. Shift from treating automated scoring as an **isolated prediction task** to situating it within the broader **stakeholder-centered assessment lifecycle**.

Stakeholder-Centric Interpretability Principles. The FGTI principles are derived from the stakeholder needs and benefits.

Method. AnalyticScore is a reference framework implementing FGTI

Evaluations. AnalyticScore achieves high scoring accuracy and aligns well with human judgment

Interpretable automated scoring is **not just about scoring**.
It should be designed to support the **full assessment lifecycle**.

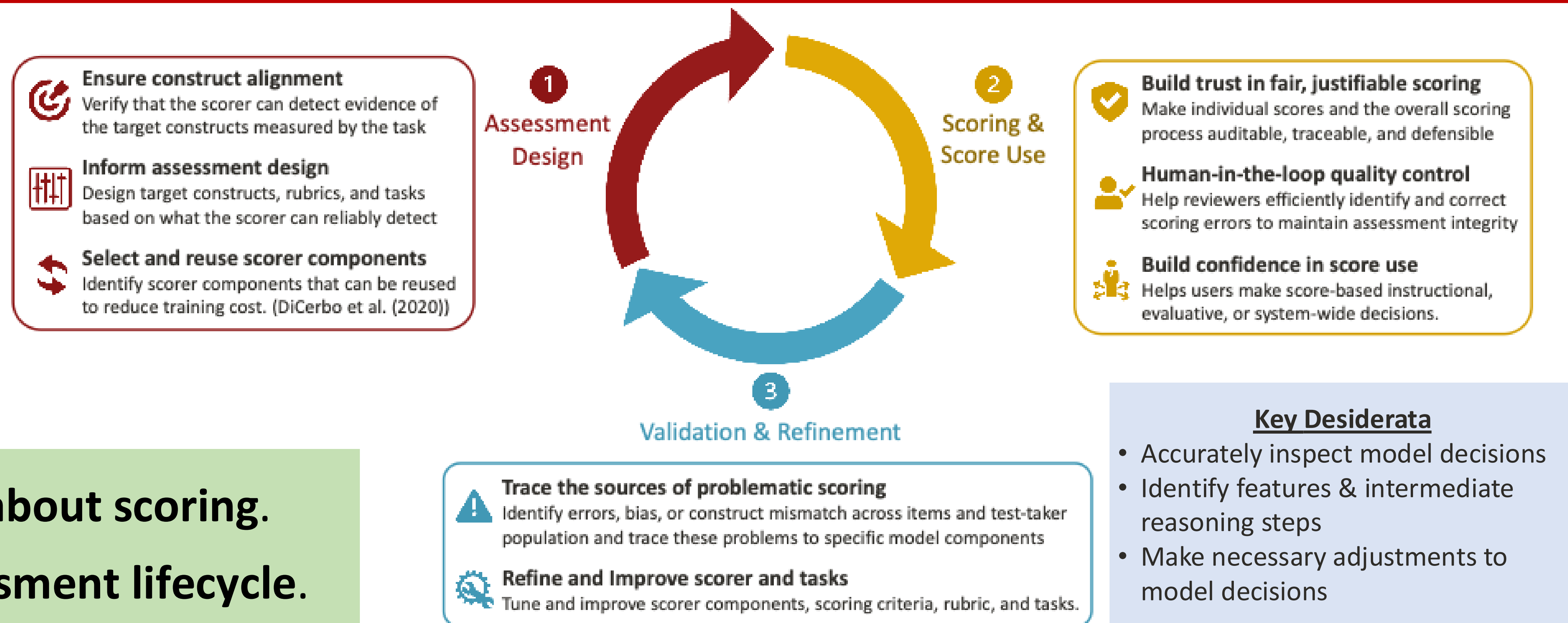
Key Takeaways

Design of interpretable AI should be grounded in **stakeholder needs**.

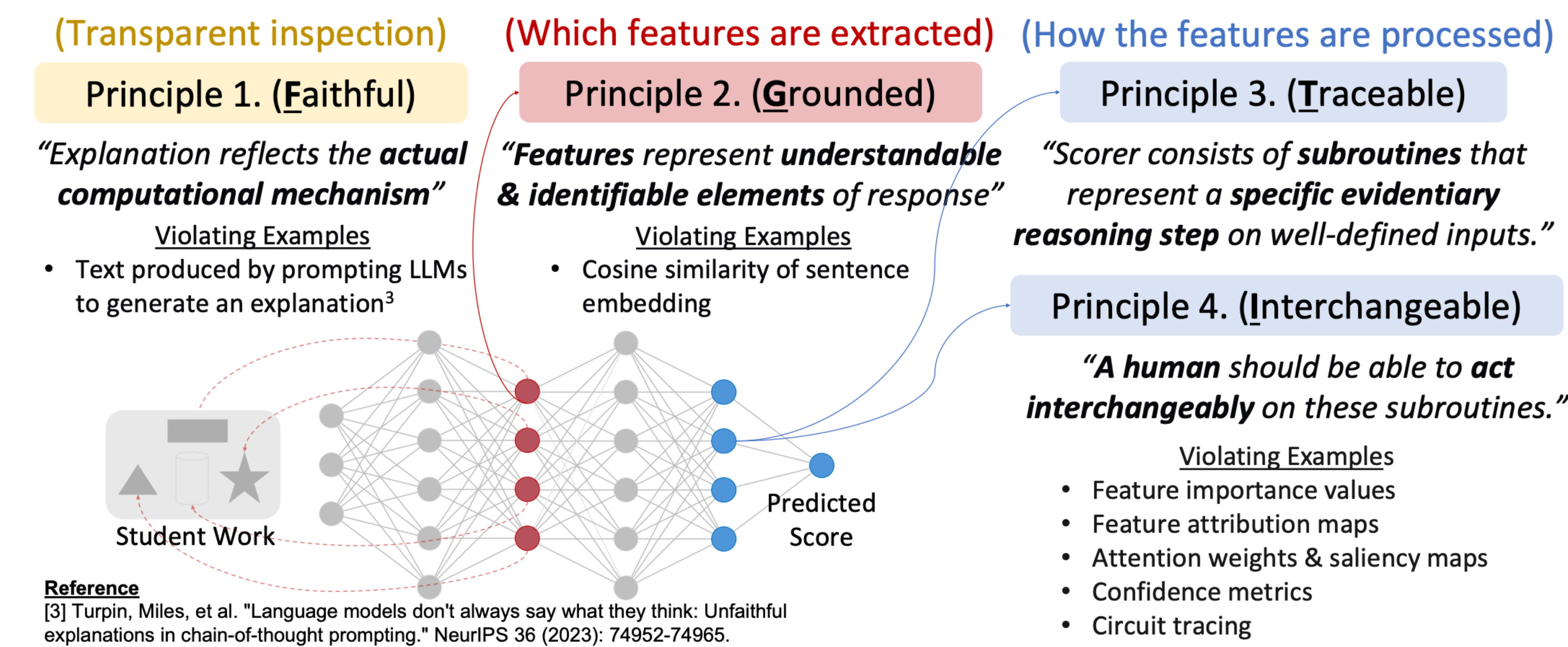
We derive the “**FGTI**” principles of interpretable automated scoring by examining the needs of the **assessment stakeholders**.

AnalyticScore embodies the FGTI. It is accurate and human-aligned.

Stakeholder Needs & Benefits of Interpretable Automated Scoring

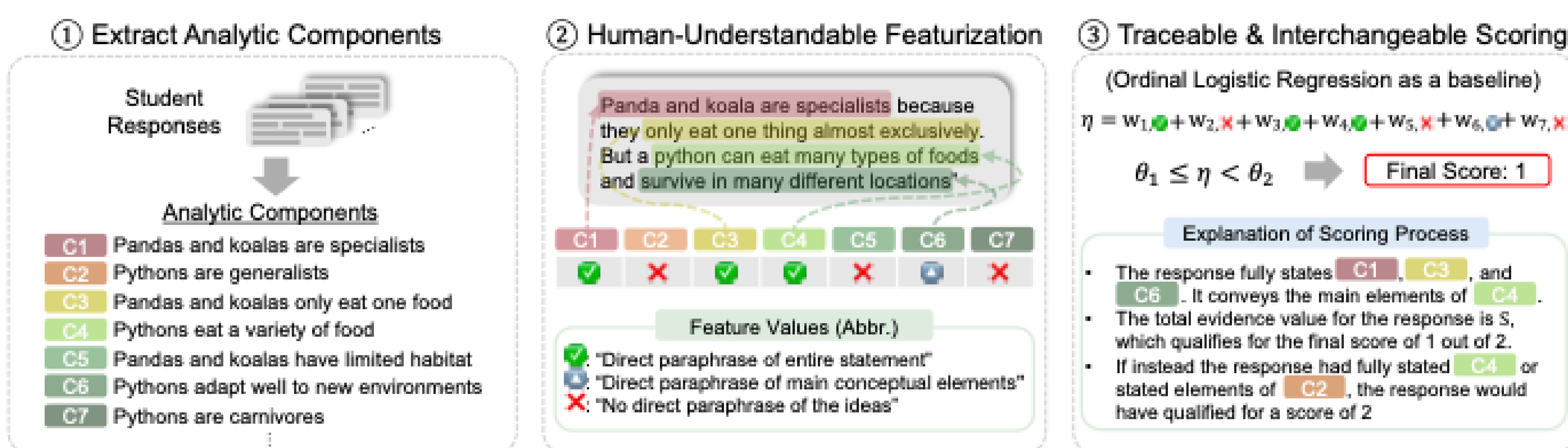


The FGTI Principles of Interpretable Automated Scoring



AnalyticScore: Reference Framework for the FGTI Principles

Q: Explain how pandas in China and koalas in Australia are similar, and how they both are different from pythons.



Analytic Components: “Explicitly identifiable elements of student responses” (Groundedness)

- Can be reviewed and modified by assessment developers during assessment design & refinement
- For constructed response scoring, analytic components are propositions

Distilling to Open Source. Supervised Fine-Tuning on 10,000 featurization examples from a larger LLM featurizer to train an open-source featurizer to avoid linearly growing cost.

Evaluation Dataset

ASAP-SAS Dataset: Largest public constructed-response scoring dataset with complex response structures. 10 items, 3 assessment areas.

Evaluation of AnalyticScore

Scoring Accuracy Experiments

- Despite limited architectural flexibilities, AnalyticScore outperforms several baseline scorers including proprietary LLMs and achieves accuracy close to black-box SOTA.
- Distillation substantially improves accuracy to the level of proprietary LLM featurizers.

	All Avg.	Sci Avg.	R(Inf) Avg.	R(Lit) Avg.
Human Scorers	0.88±0.02	0.93±0.01	0.79±0.03	0.91±0.05
ANALYTICSCORE				
w/ GPT-4.1-mini	0.72±0.03	0.78±0.03	0.68±0.06	0.60±0.01
w/ Llama-3.1-8b Inst + Distillation	0.58±0.03	0.58±0.04	0.63±0.05	0.48±0.04
	0.71±0.03	0.77±0.03	0.68±0.06	0.60±0.01
Few-Shot				
GPT-4.1	0.63±0.04	0.67±0.02	0.68±0.04	0.45±0.12
Supervised LLM				
BERT	0.75±0.02	0.79±0.02	0.74±0.05	0.69±0.01
DeBERTa	0.76±0.03	0.81±0.03	0.72±0.04	0.67±0.04
Llama-3.1-8b Inst.	0.76±0.02	0.79±0.02	0.75±0.03	0.69±0.02
w/ rubric	0.78±0.02	0.82±0.02	0.76±0.05	0.68±0.04
Llama-3.1-8b	0.76±0.02	0.80±0.02	0.76±0.03	0.67±0.02
Baseline				
AutoSAS	0.56±0.04	0.57±0.04	0.65±0.06	0.41±0.04
ASRRN	0.59±0.02	0.59±0.04	0.63±0.04	0.55±0.04
NAM	0.56±0.05	0.64±0.02	0.52±0.13	0.40±0.02

Table 1. Scoring accuracy measured by QWK against gold standard scoring labels

Featurization Alignment Experiments

7 annotators independently performed AnalyticScore’s featurization.

Question 1: Is AnalyticScore’s featurization task clear enough to yield human agreement?

- Human raters agree well on the featurization task (Table 2)

Question 2: Does AnalyticScore’s featurization align with human judgment?

- Distilled featurizer achieves high agreement with aggregate human features
- Further study should reduce the ambiguity of the “partial paraphrase” category

Assessment Area	Krippendorff’s α
Science	(0.718, 0.723)
Reading (Informational Text)	(0.696, 0.700)
Reading (Literature)	(0.672, 0.680)

Table 2. Inter-rater reliability among human raters on AnalyticScore’s featurization task

Assessment Area	Featurizer Model	QWK	Label-wise F1		
			✓	⬆	✗
Science	GPT-4.1-mini	(0.89, 0.89)	(0.83, 0.84)	(0.20, 0.21)	(0.96, 0.96)
	o4-mini	(0.94, 0.95)	(0.93, 0.93)	(0.49, 0.51)	(0.98, 0.98)
Reading (Informational Text)	GPT-4.1-mini	(0.71, 0.72)	(0.68, 0.69)	(0.54, 0.55)	(0.87, 0.88)
	o4-mini	(0.81, 0.82)	(0.73, 0.74)	(0.69, 0.70)	(0.94, 0.94)
Reading (Literature)	GPT-4.1-mini	(0.52, 0.54)	(0.44, 0.46)	(0.67, 0.69)	(0.95, 0.95)
	o4-mini	(0.49, 0.51)	(0.49, 0.51)	(0.10, 0.12)	(0.92, 0.92)
	Distilled	(0.81, 0.82)	(0.86, 0.87)	(0.17, 0.19)	(0.92, 0.92)

Table 3. LLM featurizers vs aggregate human features